

02.11.2015 (E)

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

Semester: July 2015 - November 2015

Max. Marks: 60**Duration: 3hrs****End Semester Exam**

Class: SE

Semester: III

Name of the Course: OOPM

Branch: COMP

Course Code: UCEC302

Instructions:

- (1) All Questions are Compulsory
- (2) Make suitable assumptions if required and state them clearly
- (3) All programs and program segments should be strictly in JAVA.

Question No.		Max. Marks
Q1(a)	Write a recursive method to calculate factorial of an Integer number. OR Write a program to accept an Integer number from command line and check whether it is Armstrong number or not.	[05]
Q1(b)	Explain how java is platform independent in brief.	[05]
Q1(c)	Identify classes and their attributes and draw the relationships specified by following problem. "Bank maintains two kinds of accounts for customer, savings account and current account. Savings account provides compound interest and withdrawal facility. Current account provides cheque book facility but no interest".	[05]
Q1(d)	Elaborate the use of keyword super with example.	[05]
Q2(a)	Compare and contrast between method overloading and method overriding with example. OR What is an abstract class? How it is different from interface explain with an example.	[10]
Q2(b)	Write a program to display area of square and rectangle. Make use of interface to define templates of methods to be there in the derive classes.	[10]
Q3(a)	Write detailed note on following exception handling terms. a) try-catch b) finally	[10]

	c) catch multiple exception d) throwing exception OR Write a program in java to accept the values of a, b, c, d and calculate and display $((a*b) + (b*c)) / (b*d)$. Create user defined exception to display proper message when value of (b*d) is ZERO .	
Q3(b)	What is a package? Explain with example the steps to create package and add a class or an interface.	[10]
Q4 (a)	What is a thread? Elaborate on life cycle of a Thread and what are the different ways to create thread in java. OR What are synchronization of thread in java? Write a program to synchronize two threads	[10]
Q4 (b)	What is an applet? Write a program to pass parameter in an applet. OR Write an applet program for smiling face. 	[10]
Q5 (a)	Write short notes on (Any Two). a) toString() method b) wrapper classes in java c) System.arraycopy() method OR Elaborate on the usage of final keyword, finally and finalize() method with example.	[10]
Q5 (b)	Write a program to create vector objects with student names. Program should perform following operations based on choice: a) Add student name: to add new student name in the vector. b) Remove student name: removes student name if already exists else display appropriate message. c) Display: Display contents of vector.	[10]